

CHE 301 Hw 4

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Abstract

This report presents a comprehensive computational analysis of thermodynamic properties and flow behavior for fluids under varying conditions. Utilizing the CHE Calculator and Excel-based modeling, we evaluate the thermal expansion coefficient (β), enthalpy, and entropy through Akima spline interpolation and ideal gas relations. Problem A focuses on deriving β from interpolated density data, followed by enthalpy and entropy calculations at constant temperature and pressure. Problem B explores dynamic mass and energy balances during venting, including isentropic flow conditions, critical mass flow rate, and pipe diameter estimation. The integration of numerical methods and thermodynamic principles enables accurate prediction of system behavior under transient and steady-state conditions.

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1 Background/Assumptions

We first need extensive background knowledge of the CHE calculator to complete this assignment. Knowledge of the akima splines function was key in solving problem A, as well as knowledge of the equations of the thermal expansion coefficient, enthalpy, and entropy. For problem B, knowledge about unsteady mass balances, unsteady energy balance, and isentropic relations are key. We need form relations between these equations to determine the properties needed in the solution. We also use the ideal gas law extensively throughout the problem.

2 Problem A

The first step in solving this problem is to identify the equations needed. For β , we use:

$$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_P$$

For enthalpy, we use:

$$dH = C_p dT + (1 - \beta T) V dP$$

For entropy, we use:

$$dS = C_p \frac{dT}{T} - \beta V dP$$

2.1 Solving for β

From the Z values provided, we can determine the density. We utilize the `ATMtoBAR()` function in the CHE Calculator to convert all pressures to bar, while the temperature remains the same. The `Density()` function in the CHE Calculator can then be used to determine the density by inserting the temperature in °C, pressure in bar, and Z values from the table. The density is obtained in units of kmol/m³.

The corresponding function used in Excel is:

```
=Density(C$3,$B22,C4)
```

This formula was dragged down to compute all density values for the different conditions as seen in Figure 1 below. To find β , we need to compute the density values for all temperatures at 700 bar. For this, the `fAkima()` function in the CHE Calculator is used. The x -values correspond to the pressures in bar, the f -values correspond to the density values for the column of the temperature of interest, `job` is set to 1 since we are interpolating, and X_0 is set to 700 (for 700 bar). In the Excel sheet, the pressure value cell was referenced to allow flexibility in changing it.

Density Table						
	Temp					
P (bar)	0	25	50	75	100	
0	0	0	0	0	0	
1.01325	0.044615	0.040874	0.037712	0.035004	0.032659	
3.03975	0.133845	0.122622	0.113136	0.105012	0.097976	
5.06625	0.223075	0.20437	0.188559	0.175019	0.163294	
10.1325	0.44615	0.40874	0.377119	0.350039	0.326587	
30.3975	1.338451	1.226221	1.131357	1.050116	0.979761	
50.6625	2.230752	2.043702	1.885594	1.750193	1.632935	
101.325	4.461503	4.087404	3.771189	3.500387	3.265871	
202.65	8.923007	8.174809	7.542377	7.000773	6.531741	
303.975	13.38451	12.26221	11.31357	10.50116	9.797612	
405.3	17.84601	16.34962	15.08475	14.00155	13.06348	
607.95	26.76902	24.52443	22.62713	21.00232	19.59522	
810.6	35.69203	32.69924	30.16951	28.00309	26.12697	
1013.25	44.61503	40.87404	37.71189	35.00387	32.65871	

Figure 1: Density Table with All Pressure and Temperatures

	B	C	D	E	F	G	H
37	Density Values						
38		T C					
39	P bar	0	25	50	75	100	60
40	700	30.82213	28.23768	26.05312	24.18229	22.56215	25.274

Figure 2: Density Table with Constant Pressure

The function in the Excel formula bar is:

```
=fAkima(B22:B35,C22:C35,1,B40)
```

This procedure was repeated for all temperatures. Next, using the new table of density values, Akima spline interpolation was applied to determine the density at 60 °C and 700 bar. The new x -values correspond to temperatures from 0 to 100 °C, the f -values correspond to the densities at 700 bar, job remains 1, and $X_0 = 60$. The resulting density is found to be:

$$\rho = 25.274 \text{ kmol/m}^3$$

as seen in Figure 2. From this information, β can be calculated using

$$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_P$$

. Akima splines are again utilized to compute the required derivative numerically. The x -values are again the temperatures, the f -values are the densities, the job changes to 3 since we are computing a derivative, and $X_0 = 60$. In my excel sheet, the full function for beta is

$$=-1/H40*fAkima(C39:H39,C40:H40,3,H39)$$

where H40 is the density at 700 bar and 60 °C. From this we found

$$\beta = .00297 \text{ K}^{-1}$$

2.2 Solving for Enthalpy

In order to compute Enthalpy, we need to manipulate some equations. When computing enthalpy, we will keep temperature constant in order to cancel the term $C_p dT$. Thus this equation:

$$dH = C_p dT + (1 - \beta T)V dP$$

Turns into

$$dH = (1 - \beta T)V dP$$

We can do the same with this equation

$$H = H_f + \int C_p dT$$

Where $C_p dT$ will cancel. By combining these equations, we can get

$$H = H^{\text{IG}} + \int_0^{700} (1 - \beta T)V dp$$

Since the temperature is held constant for the computation of enthalpy, we must once again utilize the **fAkima** function to create a row of density values corresponding to all pressures at a temperature of 60 °C.

The function used in Excel is shown below:

$$=fAkima(\$C\$21:\$G\$21, C22:G22, 1, \$I\$21)$$

where:

- **xvalues** are the temperatures,
- **fvalues** are the densities at the specified pressures,
- **job** is set to 1 (to evaluate the function), and
- **X₀** is 60 °C.

This formula was then dragged downward to compute the density values for all given pressures as seen in Figure 3.

Next I created a column titled $\frac{1}{\rho}(1 - \beta T)$, where β is the value we found and T is 60. I then dragged it down to the last given pressure value. To compute the enthalpy value, the **HIG()** function in Excel was utilized with a temperature of 60 °C. For the **molfrac** term, an absolute reference **comp1** was used to access

	H	I	J
19		Density for different pressure	
20		Temp C	
21	P (bar)	60	V m³/Kmol
22	0	0	0
23	1.01325	0.036584117	27.33426617
24	3.03975	0.109752352	9.111422056
25	5.06625	0.182920587	5.466853234
26	10.1325	0.365841173	2.733426617
27	30.3975	1.097523519	0.911142206
28	50.6625	1.829205865	0.546685323
29	101.325	3.658411731	0.273342662
30	202.65	7.316823461	0.136671331
31	303.975	10.97523519	0.091114221
32	405.3	14.63364692	0.068335665
33	607.95	21.95047038	0.04555711
34	810.6	29.26729385	0.034167833
35	1013.25	36.58411731	0.027334266

Figure 3: Density and Volume Table with Constant Temp

the component entered into the CHE Calculator. Next, we added the integral term, which was computed using the `fAkima()` function. For this function, the `xvalues` correspond to the pressures, the `fvalues` are the densities at 60 °C, the `Job` parameter was set to 5 to perform an integration, the lower limit was 0, and the upper limit was 700 bar. The complete Excel formula entered into the function bar is shown below:

```
=HIG(comp1,I21) + fAkima(B23:B35,L23:L35,5,,0,B40))/1000
```

The factor of $\frac{1}{1000}$ is used to convert from KJ/kmol to kJ/mol. From this we obtained a value of 1.023 kJ/mol.

2.3 Solving for Entropy

For entropy, we use the same concept from the enthalpy calculations. The equation we are using is

$$S = S^{\text{IG}} - \int_0^{700} \beta V dp$$

I made a column with $-\beta V$, and dragged it down to the last pressure values seen in Figure 4. Then to find the entropy, I first used the `SIG()` function in Excel with a temperature of 60 °C and 700 bar. For the `molfrac` term, an absolute reference `comp1` was used to access the component in the DataBase workbook. For the integral term, For this function, the `xvalues` correspond to the pressures in bar, the `fvalues` is the new column $-\beta V$, the `Job` parameter was set to 5 to perform an integration, the lower limit was 0, and the upper limit was 700 bar. The function in excel is

	L	M
19		
20		
21	V(1-BetaT)-beta*V	
22	0	0
23	0.265774	-0.000108745
24	0.088591	-0.000326235
25	0.053155	-0.000543725
26	0.026577	-0.00108745
27	0.008859	-0.003262351
28	0.005315	-0.005437251
29	0.002658	-0.010874503
30	0.001329	-0.021749006
31	0.000886	-0.032623508
32	0.000664	-0.043498011
33	0.000443	-0.065247017
34	0.000332	-0.086996023
35	0.000266	-0.108745028
36		

Figure 4: Integral Prep Tables

```
=SIG(comp1,H39,B40)+fAkima(B23:B35,M23:M35,5,,0,B40)
```

From this we got a value of -77.41 KJ/kmolK

	J	K
40	Beta K ⁻¹	0.002972466
41	Hr	2.174592145
42	Sr	-26.29413465
43		
44	hig	1021.017626
45	sig	-51.12129273
46		
47	H KJ/mol	1.023192218
48	S (KJ/KmolK)	-77.41542738
49		

Figure 5: Final Values

3 Problem B

3.1 Question 1

In order to find the initial mass of ethylene in the tank, we utilize ideal gas properties as specified in the problem. From the ideal gas law we can see:

$$\begin{aligned}\frac{PV}{RT} &= n \\ P &= 200 \text{ bar} \quad T = 400 \text{ K} \\ V &= 20 \text{ m}^3 \quad R = 8.314 \text{ m}^3 \cdot \text{Pa}/(\text{K} \cdot \text{mol}) \times \frac{1 \times 10^5 \text{ Pa}}{1 \text{ bar}} \\ n &= \frac{PV}{RT} \quad (\text{in moles}) \\ \text{Mass} &= n \times MW \times \frac{1 \text{ kg}}{1000 \text{ g}}\end{aligned}$$

Where n is moles, MW is the molecular weight of ethylene. Here we had to change our gas constant from pascal to bar, and gram to kilograms. From this, our answer for the initial mass of ethylene in the tank is 3374 Kg. We did this calculation through excel.

```
=C6*C28/C4/C16*molw2*100
```

is what we put in the function bar, Where C6 is pressure, C28 is volume, C4 is temperature, C16 is the R constant, molw2 is the molecular weight of ethylene, and 100 is the conversion factor.

3.2 Question 2

This question is asking us to determine the pressure and temperature of the narrow section of the pipe once the critical mass flow rate is reached at 5000 kg/hr. We are given that this critical mass flow rate corresponds to the speed of sound. The venting process is also assumed to be reversible and adiabatic and the ethylene inside an ideal gas.

Starting with the unsteady-state mass balance, because the only flow of mass present is going out of the system, we get

$$\frac{dM}{dt} = -m$$

Combining this with the entropy balance:

$$\frac{d(MS)}{dt} = -mS + S_{\text{gen}} + \frac{Q}{T}$$

No entropy is being generated because the system is reversible, and there is no heat transfer because the system is adiabatic, so:

$$\frac{d(MS)}{dt} = -mS$$

Combining this equation with the mass balance equation, we get:

$$d(MS) = -mS dt = dMS$$

Then, taking the integral,

$$M_0S_0 = MS_1 = (M_0 - M)S_1$$

Which simplifies to

$$S_0 = S_1$$

So, the entropy in the tank before the valve is opened is the same as the entropy at the critical mass flow.

For the energy balance of the system:

$$\frac{dU_{\text{tot}}}{dt} = - \left(\left[H_1 + \frac{1}{2}v_1^2 + gh_2 \right] \rho_1 v_1 A_1 \right) + \dot{Q} + \dot{W}_s$$

Assuming potential energy is negligible, there is no heat transfer (adiabatic), and no shaft work done, the equation simplifies to:

$$\frac{dU_{\text{tot}}}{dt} = - \left(\left[H_1 + \frac{1}{2}v_1^2 \right] m \right)$$

Considering the following equation for energy:

$$dU = T dS - P dV$$

We from the entropy balance that entropy is constant, so $dS = 0$.

$$dU = -P dV$$

Considering the enthalpy equation:

$$H = U + PV$$

$$dH = dU + VdP + PdV$$

And using the identity of dU:

$$dU = TdS - PdV$$

$$dH = (TdS - PdV) + PdV + VdP$$

Because entropy is constant and the PdV terms cancel out,

$$dH = VdP$$

$$H = PV$$

Using the Bernoulli equation of fluid dynamics (assuming C is a constant):

$$\frac{1}{2}v^2 + PV + gz = \text{constant}$$

The potential energy effects in this system are negligible, so

$$\frac{1}{2}v^2 + PV = C$$

And because $H = PV$,

$$\frac{1}{2}v^2 + H = C$$

When the valve is shut at the beginning of the venting process, when enthalpy is H_0 , the velocity of the gas is $v = 0$, which gives us:

$$H_0 = C$$

Therefore,

$$H_0 = H_1 + \frac{1}{2}v^2$$

Knowing these relationships, we find H_0 , H_1 , S_0 , and S_1 using the CHE Calculator by using the functions =HIG (inputting ethylene and temperature) and =SIG (inputting ethylene, temperature, and pressure). Because we do not yet know the true temperature T_1 and pressure P_1 , we must input an initial guess for both, which we used 100°C and 100 bar. To find the speed of the fluid, the speed of sound, we use the equation:

$$v = \sqrt{\frac{\gamma RT}{M_w}}$$

Heat capacities C_p and C_v were found using the =CpIG and =CvIG functions, inputting composition ethylene and temperature. The output values were $C_p=50.73$ kJ/kmol K and $C_v=42.42$ kJ/kmol K. The ratio of the heat capacities give us γ :

$$\gamma = \frac{C_p}{C_v} = 1.196$$

To make sure units stayed consistent, the R value we used was the relationship between C_p and C_v for ideal gases:

$$R = C_p - C_v = 8.314 \text{ kJ/kmol K}$$

Inputting these values into the speed of sound equation, we get

$$v = 359.88 \text{ m/s}$$

Now that we have the velocity of the fluid, we can plug it into the enthalpy balance:

$$0 = H_1 + \frac{1}{2}v^2 - H_0$$

With our original guess for temperature T_1 , the above balance will not equal zero. In Excel, we used the Goal Seek function to set this balance equal to zero by changing the value of T_1 . As a result, we got $T_1 = 92.233^\circ\text{C}$. However, the

entropy balance is not satisfied, yet, as $S_0 - S_1 \neq 0$. So, we used Goal Seek to set the subtraction of the entropies equal to 0 by changing the pressure P_1 . As a result, we got a pressure of $P_1 = 112.98$ bar.

So, the temperature and pressure of the gas in the narrow section of the pipe at critical mass flow is 92.233 °C and 112.98 bar, respectively.

3.3 Question 3

The diameter of the narrow section of the venting pipe can be determined by finding its cross-sectional area, first. The relationship between area and mass flow rate is:

$$\dot{m} = A\rho v$$
$$A = \frac{\dot{m}}{\rho v}$$

The mass flow rate at the critical point was given as 5000 kg/hr, which was converted to kilograms per second by dividing 3600 seconds/hour. The molar flow was determined by dividing this value by the molar mass of ethylene.

$$\left(\frac{5000 \text{ kg}}{1 \text{ hr}}\right) \left(\frac{1 \text{ hr}}{3600 \text{ s}}\right) \left(\frac{1 \text{ kmol}}{28.013 \text{ kg}}\right) = 0.0495 \text{ kmol/s}$$

The density was found using the =density function in the CHE Calculator, inputting our T_1 and P_1 , which gave an output of 3.719 kmol/m³. The velocity is the speed of sound found in the previous question: $v = 359.88$ m/s. Combining these in the area equation:

$$A = 3.6989 \times 10^{-5} \text{ m}^2$$

Assuming the pipe is a cylinder, the cross-section is a circle with the following equation relating diameter and area:

$$d = 2\sqrt{\frac{A}{\pi}} = 6.86 \text{ mm}$$

The diameter of the pipe at the narrow section is 6.86 mm.

3.4 Question 4

This problem asks what the mass of ethylene is in the tank when the critical condition is reached. This problem has an approach very similar to Question 1.

From the ideal gas law we can see:

$$\frac{PV}{RT} = n$$

$$P = 112.9843 \text{ bar} \quad T = 365.3829 \text{ K}$$

$$V = 20 \text{ m}^3 \quad R = 8.314 \text{ m}^3 \cdot \text{Pa}/(\text{K} \cdot \text{mol}) \times \frac{1 \times 10^5 \text{ Pa}}{1 \text{ bar}}$$

$$n = \frac{PV}{RT} \quad (\text{in moles})$$

$$\text{Mass} = n \times MW \times \frac{1 \text{ kg}}{1000 \text{ g}}$$

The only values that changed from Question 1 is the pressure and temperature, which we found when solving Question 2. From this, we determined that the mass of ethylene in the tank at the critical point is 2086.7 Kg. The excel functions we used are very similar to Question 1.

```
=D6*D28/C16/D4*100*molw2
```

3.5 Question 5

For finding the mass of ethylene in the tank when the pressure gauge reads 2300 PSI, the same approach as in Questions 1 and 4 is utilized, but with additional steps. First, we convert the pressure of 2300 PSI to bar using the `PSItoBAR()` function in the CHE calculator, obtaining a pressure of 158.58 bar. An initial temperature of 100 °C was assumed as a starting guess, which will later be adjusted using Goal Seek.

Next, the molar entropy in kJ/kmol·K is calculated using the `SIG()` function.

```
=SIG(comp2,E5,E6)
```

where `comp2` is the composition of ethylene, `E5` is the temperature that was guessed, and `E6` is 158.58 bar. Since this process is isentropic, the entropy of the previous stream must be equal to that of the new stream. Therefore, we use Goal Seek to solve for the temperature that satisfies the entropy balance, by setting the entropy difference to zero and changing the temperature.

As a result, the final temperature of this new stream is found to be 112.6813 °C, which corresponds to 385.8313 K.

The rest of the problem has the same process Question 1 and 4.

$$\frac{PV}{RT} = n$$

$$P = 158.5789 \text{ bar} \quad T = 385.8313 \text{ K}$$

$$V = 20 \text{ m}^3 \quad R = 8.314 \text{ J} \cdot \text{Pa} / (\text{K} \cdot \text{mol}) \times \frac{1 \times 10^5 \text{ Pa}}{1 \text{ bar}}$$

$$n = \frac{PV}{RT} \quad (\text{in moles})$$

$$\text{Mass} = n \times MW \times \frac{1 \text{ kg}}{1000 \text{ g}}$$

In excel, the function bar reads

```
=E6*E28/E16/E4*100*mo1w2
```

where E6 is the pressure, E28 is the volume, E16 is the gas constant, and E4 is the temperature. Thus, the mass of ethylene in the tank at 2300 PSI is 27733.57 Kg.

	A	B	C	D	E	F	G	H
1								
2								
3			S0	S1	S2 at 2300 PSI		Initial mass of ethylene (kg)	3374.12065
4		Temp K	400	365.3829	385.83			
5		Temp C	126.85	92.23286	112.68		Area of pipe m ²	3.6989E-05
6		Pressure (Bar)	200	112.9843	158.579		diameter of pipe (mm)	6.86264934
7								
8		Mass flow kg/s		1.38888889			Mass of ethylene at crit pt (kg)	2086.70245
9							mass of ethylene at 2300 PSI (kg)	2773.56531
10								
11		Enthalpy KJ/mol	57323.97357	55507.2379	56565.9			
12		entropy KJ/mol/K	-82.7063037	-82.706304	-82.706			
13		density kmol/m ³	6.013617752	3.71908186	4.94326			
14		density kg/m ³	168.7060324	104.335122	138.678			
15								
16		R (KJ/KmolK)	8.314462618	8.31446262	8.31446			
17		Cp	54.20603068	50.7302983	52.8018			
18		Cv	45.89156806	42.4158357	44.4873			
19		gamma	1.181176259	1.19602261	1.1869			
20								
21		Speed of sound m/s		359.884776	368.404			
22		kinetic energy J/Kg		64758.5261	67860.8			
23		KE KJ/mol		1816.73569	1903.77			
24								
25		enthalpy balance	6.91216E-09					
26		entropy balance	-2.615E-08	2.1844E-09				
27								
28		volume of tank m ³	20	20	20			
29								
30								

Figure 6: Part B Full Excel Sheet

4 Conclusion

Through methodical application of thermodynamic equations and computational tools, this assignment successfully quantifies key properties of fluids across a range of pressures and temperatures. Akima spline interpolation proved essential for estimating derivatives and integrals within non-linear datasets. The use of reversible and adiabatic assumptions in Problem B allowed for simplification of entropy and energy balances, leading to precise determination of flow parameters such as velocity and pipe diameter. Overall, this work demonstrates the power of combining theoretical models with data-driven computation to solve complex chemical engineering problems with clarity and reproducibility.